

word. For instance, if they displayed the word 'Go', he would touch the same word that appeared on the makeshift keyboard with his left toe. The staff would then display a chart with images of six different places such as 'Home', 'School', 'Bathroom', 'Park' and so on.

While this may appear to be easy to an onlooker, the fact is that those affected with cerebral palsy can only identify the pattern of the process, and may not necessarily comprehend the written word.

However, as Sachar grew proficient at correctly associating the written word with the image, it became difficult to conduct the project on paper. As the cut-outs increased in number, D'Souza realised she would need to get a special keyboard to continue the phase-wise expansion of Sachar's learning process.

It was then that D'Souza approached Prof Gaur G Ray, a faculty member at the Industrial Design Centre (IDC) at IIT Bombay, a premier design institute in India.

From Paper To PC

A physiology and bio-mechanics expert, Ray has been teaching at IIT Bombay for the past 24 years. His area of specialisation includes ergonomics—developing designs that are physiologically, bio-mechanically and psycho-physiologically suited to the human body.

Ray has been involved with various spastic associations (including Happy Hours centre) in India as part of his work. When D'Souza inquired about the feasibility of making the special keyboard, Ray took up the challenge.

Ray summarised the plan and assigned the project to his students in the biomedical engineering department—he believed that the ideas from those studying human anatomy and engineering would lead to the best prototype. Meanwhile, he applied to the Ministry of Human Resource Development (MHRD) for funding the project in 2003, and it was duly sanctioned.

It's Different

Ray's students developed a keyboard that had the aforementioned six basic functions—one key for each function. It was customised to meet Sachar's needs. They also designed a software in Visual Basic that linked the words to the images.

When Sachar hit the button labeled 'Drink', the screen would display six pictures of—'Water', 'Tea', 'Coffee', 'Coke', 'Milk' and 'Juice'. Next, he'd choose any one by pressing on the keyboard, the same digit assigned to the picture on the screen. Say, if he wanted water, which was tagged with '1' on the screen, he would press the digit '1' on the keyboard. This would bring up another screen with an image of a glass of water. This helped Sachar communicate his needs to the person in his room.

Specifics

The keyboard had digits from one to 12, and keys such as 'Back' and



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Prof G G Ray
Industrial Design
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'Forward' for ease of navigation. It was made of a metal frame, and the keys were industrial heavy-duty switches, the tops of which were removed and fitted with an acrylic material. They were then moulded so as to resemble keys on a typical keyboard using a method called 'baceum forming'.

The students also designed the seating arrangement, taking into account the workspace Sachar needed—the monitor had to be at a certain angle, keyboard had to be positioned such that Sachar could operate it with his left foot.

They used a 21-inch monitor. Even the keys were unusually large, and spaced wide apart. As Ray says, "We had to design in terms of inches and not millimetres. Moreover, it turned out to be a very low-cost design."

Accolades And Upgrades

In 2004, this unique keyboard and software package received the 'Best Computer Application Award' at the International Congress on Working with Computers in Malaysia.

Over time, Ray improvised the entire package, and continues to do so. "Based on the feedback we received from the staff at Happy Hours Centre, we made the desired changes in the package. Initially, what was meant to be a learning aid developed into an effective means of communication for Sachar," says Ray.

Sachar has been using this set-up for two years now, and other kids affected with cerebral palsy are also using it. But since Sachar operates the keyboard with his foot, the force with which he hits the keys renders the keyboard unusable after a period of time.

Hence, Ray is working on what he calls an intelligent plate—a smooth rectangular board made of stronger material. He is trying to make it touch-sensitive and also deploy a sensor that will detect which part of the plate is being hit so that the software can, in response, flash relevant images.

Ray bemoans the lack of government initiative to popularise this project. The highly expensive state-of-the-art computers for cerebral palsy patients available abroad are still a distant dream in India. In comparison, the unique product designed by Ray makes for a practical, viable and economical option. ☐

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Photograph: Jiten Gandhi



The trainer at Happy Hours centre helps Sachar learn newer words. Inset—the special keyboard